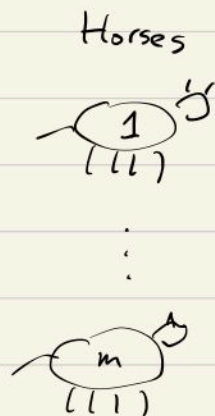


Ch 6, Gambling and Data compression

"Gambling in terms of information theory!"

↳ Horse Racing.



Winning rate (= Return per 1, if won)

p_1 } not known
 $\vdots$
 p_m

O_1 } known.
 $\vdots$
 O_m

Given

Gambler's investment (betting)

$b_1 \rightarrow (h_1, p_1, O_1)$

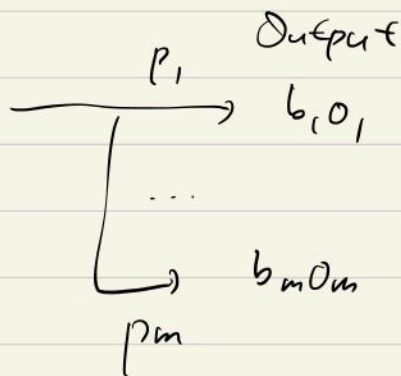
* Notation
 $(b_1, \dots, b_n) = b$
 $(p_1, \dots, p_n) = p$

b_m

Can be chosen

Note $\sum_{i=1}^n b_i = 1$
 $b_i \geq 0, \forall i$

Input
1



② Maximum (Output?)
↳ $\mathbb{E}$

Utility
(Low risk wanted)

$$\Rightarrow \mathbb{E} = \sum_{i=1}^m p_i b_i o_i$$

$$\leq \left(\sum_{i=1}^m b_i \right) \max_i (p_i o_i) = \max_i (p_i o_i).$$

* Convex function might be used:

$$\sum_i p_i f(x_i) \xrightarrow{\downarrow} f\left(\sum_i p_i x_i\right) \text{ as } \max_i \|x_i - x_j\| \rightarrow 0$$

f is convex

$$\} f = \log$$

Def, (Doubling rate)

$$W(b; p) = \mathbb{E}[\log S(x)] = \sum_{x=1}^m p_x \log(b x o_x)$$

Notation // X_i : i.i.d v.v. with $\mathbb{P}(X_i = b_i o_i) = p_i, \frac{1}{i}$.

$S(X_i) := b(X_i) o(X_i) \rightarrow$ Return at i^{th} race

$S_n(X_i) := \prod_{i=1}^n S(X_i) =$ Wealth after n races.

($\rightarrow$ Underlying assumption ; p_i & o_i is not change)

Observation //

$\frac{1}{n} \log S_n = \frac{1}{n} \sum_{i=1}^n \log S(X_i) \xrightarrow{\text{LLN}} \mathbb{E}[\log S(X)]$ in probability

$$\Rightarrow S_n = 2^{(n \underbrace{\mathbb{E}(\log(S(X)))}_{= W(b;p)})}$$

$W > 0$; profit

$W = 0$; same money

$W < 0$; lose money.

Thm 6.1.1. // $S_n = 2^{nW(b;p)}$

Prop. $W^*(p) := \max_b W(b, p)$ is achieved if $b = p$

We want to

$$\begin{cases} \text{Maximize } \sum p_i \ln(b_i o_i) \\ \text{subject to } \sum b_i = 1 \\ (0 \leq b_i \leq 1) \end{cases}$$

Lagrange
multiplier
method

$$\nabla_b f \parallel \nabla_b g$$

$$\iff \left(\frac{p_1}{b_1}, \dots, \frac{p_n}{b_n} \right) \parallel (1, 1, \dots, 1)$$

$$\implies \frac{p_1}{b_1} = \frac{p_2}{b_2} = \dots = \frac{p_n}{b_n} = \frac{p_1 + p_2 + \dots + p_n}{b_1 + b_2 + \dots + b_n} = 1$$

$$\implies p_i = b_i \quad \forall_i \implies p = b \quad \square$$

Thm 6.1.2.

$$W^*(p) = \sum_i p_i \log o_i - H(p)$$

$$\left. \frac{d}{db} \sum_i p_i \log(b_i o_i) \right|_{b=p} = \sum_i p_i \log o_i - \underbrace{\sum_i p_i \log p_i}_{H(p)}$$

ex 6.1.1. Two horses $(h_1, p_1, 2), (h_2, p_2, 2) \rightarrow [2 \text{ for } 1]$

$$W^*(p) = \sum_i p_i \log \underbrace{O_i}_{=1} \bigg|_{O_i=2} - H(p) = 1 - H(p) \Rightarrow S_n = 2^{n(1-H(p))}$$

* Low Entropy $\rightarrow$ High return.

Recall, Let p, q be two pmf's. Then, relative entropy, or Kullback-Leibler distance $\rightarrow$ (KL distance)
btw p and q is

$$D(p \parallel q) = \sum_x p(x) \log \frac{p(x)}{q(x)}.$$

KL distance is not a mathematical distance ($\because$ Triangle inequality $\rightarrow$ False!)
Symmetry

but we have

$$D(p \parallel q) \geq 0, \text{ with equality iff } p \equiv q.$$

* Bookmaker's estimate (Reference estimate)

(=Bakie, 마권업자 (배당률 정하는 사람))

Bookie $\longrightarrow$ Payoff		
h_1	$p_1 \stackrel{?}{=} r_1$	$\frac{1}{r_1}$
h_2	$p_2 \stackrel{?}{=} r_2$	$\frac{1}{r_2}$
$\vdots$	$\vdots$	$\vdots$
h_m	$p_m \stackrel{?}{=} r_m$	$\frac{1}{r_m}$

* Assume $\sum_i \frac{1}{o_i} = 1$, set $r_i := \frac{1}{o_i}$, $r = (r_1, r_2, \dots, r_n)$.

Then we may regard r as a pmf, and

$$W(b, p) = \sum_i p_i \log(b_i o_i) = \sum_i p_i \log\left(\frac{b_i p_i}{p_i r_i}\right)$$

$$= D(p \| r) - D(p \| b)$$

→ Heuristic: (Distance of bookie's estimate from true distribution)

→ (Distance of gambler's estimate from true distribution) > 0

i.e. gambler have to make 'more accurate estimate'

(smaller $D(p \| \cdot)$) than bookie's, to make a profit.

Gambler's
desire

Thm 6.1.3, (Conservation Thm)

Given $(h_1, p_1, m), \dots, (h_m, p_m, m)$ (so-called m -for-1 odd),

$$W^*(p) + H(p) = \log m$$

$$\text{pf, } W^*(p) = \sum_i p_i \log_{\substack{0_i=m}} | - H(p) = \log m - H(p). \quad \square$$

[6.2.] Gambling & Side information.

- What if we have information ' $\mathcal{Y}$ ' relevant to the outcome?

$$W^*(x) = \max_{\substack{b(x) \geq 0 \\ \sum_x b(x) = 1}} \sum_x p(x) \log b(x) o(x)$$

$\downarrow \mathcal{Y}$

$$W^*(x|\mathcal{Y}) := \max_{\substack{b(x|y) \\ b(x,y) \geq 0 \\ \sum_x b(x|y) = 1}} \sum_{x,y} p(x,y) \log (b(x|y) o(x))$$

$p(x,y)$ = prob that
 x th horse win,
when y is observed,

$b(x,y)$ = bets on x ,
when y is observed,

Recall (Mutual information)

$$I(X; Y) := \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)} = H(X) - H(X|Y) \\ = H(Y) - H(Y|X) \geq 0$$

Heuristic: Reduction in the uncertainty of X due to knowledge of Y .

Thm 6.2.1.

$$\Delta W := W^*(X|Y) - W^*(X) \stackrel{\text{Thm}}{=} I(X; Y)$$

pf, $W(b, X|Y)$ is maximal if $b(x|y) = p(x|y)$.

$$W^*(X|Y) = \sum p(x,y) \log \left(\frac{b(x|y)}{p(x)} \right) \Big|_{\substack{b(x|y) \\ = p(x|y)}} \\ \left(\sum_y p(x,y) = p(x) \right) \leftarrow = \sum_x p(x) \log p(x) + \sum_{x,y} p(x,y) \log p(x|y) \\ = W^* + H(X) - H(X|Y) = W^* + I(X; Y). \quad \square$$

16.3. Dependent Horse Races and entropy

X_i : not i.i.d $\rightarrow$ Assumption: $\{X_i\}$ forms a stochastic process,
 X_i depends on $X_1, \dots, X_{i-1}$.

Ex ① (Red & Black)

Setup,

5 horses

- 52 trump cards: 26 red, 26 black.
- Bets: next color.
- 2-for-1: Return $\begin{cases} \text{twice} & \text{if bet on right color} \\ \text{nothing} & \text{else} \end{cases}$
- Discard the used card.

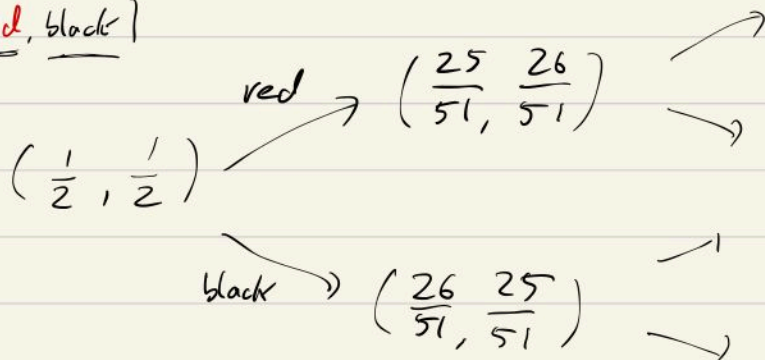
Ex ①, (Red & Black)

Setup,

51 horses

- 52 trump cards: 26 red, 26 black.
- Bets: next color.
- 2-for-1: Return $\begin{cases} \text{twice} & \text{if bets on right color} \\ \text{nothing} & \text{else} \end{cases}$
- Discard the used card.

(red, black)



Ex ②, (Guessing the alphabet)

appl ②

[Gambling & Data Compression]

In summary, 'good gambler $\rightarrow$ good data compressor!'

Setup, $X_1, \dots, X_n$: seq. of binary r.v.s we want to compress.

Bets are given by

$$b(x_{k+1} | x_1, \dots, x_k) \geq 0, \quad \sum_{x_{k+1}} b(x_{k+1} | x_1, \dots, x_k) = 1. \quad 2^{-\text{for } -1}.$$

$$\Rightarrow S_n = 2^n \prod_{k=1}^n b(x_k | x_1, \dots, x_{k-1}) = 2^n b(x_1, \dots, x_n).$$

Consider data compression

fixed, known to compressor & decompressor

$$x = x_1 x_2 \dots x_n \in \{0, 1\}^n \xleftrightarrow{\quad} C = C_1 C_2 \dots C_k \in \{0, 1\}^k$$

- Equip lexicographical order on x ,

$$(x_1 x_2 \dots x_n < y_1 y_2 \dots y_n \iff \exists k \text{ s.t. } x_i = y_i \text{ } i=1, 2, \dots, k-1 \text{ \& } x_k < y_k)$$

① Calculate $F(x(n)) := \sum_{x(n) \leq x(n)} 2^{-n} S_n(x(n)) \in [0, 1]$

② For $\underline{k} = \lceil n - \log S_n(x(n)) \rceil$, define the compressed code by

$$\frac{\lfloor 2^n F(x(n)) \rfloor}{2^n} \stackrel{\downarrow}{=} \cdot C_1 C_2 \dots C_k \mapsto C := (C_1, \dots, C_k).$$

Binary expression

③ Find $x(n)$ s.t.

$$\sum_{x(m) \leq x(n)} 2^{-n} S(x(n)) \notin [\cdot C_1 C_2 \dots C_k, \cdot C_1 C_2 \dots C_k + (\frac{1}{2})^k], \text{ but}$$

$$\sum_{x(m) \leq x(n)} 2^{-n} S(x(n)) \in \underline{\underline{[\cdot C_1 C_2 \dots C_k, \cdot C_1 C_2 \dots C_k + (\frac{1}{2})^k]}}.$$

Observation: $x(n)$ is decompression of C .

④ We can recover x from C

$\therefore$ n -bit code can be compressed by k -bits code,

$$k = \lceil n - \log S(x(n)) \rceil \implies n - k = \lfloor \log S(x(n)) \rfloor \text{ bits saved!}$$

Good at gambling $\rightarrow$ high S

$\rightarrow$ More bits saved $\rightarrow$ Good compression.